Do Language Models Use the Hops They Write? Graph-Aware Interventions on Reasoning Chains and the Limits of Contrastive Decoding

Anonymous Authors

Abstract

Chain-of-thought helps only if the answer actually depends on the steps a model writes. We study this dependence causally for multi-hop reasoning, using interventions on the model's own reasoning chain whose ground truth is known by construction. We show that the common "mask one step" test measures a controlled *direct* effect, which is provably zero for intermediate hops of bridge-entity chains, and we propose a graph-aware intervention that masks a hop together with its descendants in the chain's directed reasoning graph (DRG) and measures the *total* effect. On MuSiQue with Qwen2.5-1.5B, the graph-aware estimator separates necessary hops from inserted redundant ones with AUC 0.97 versus 0.79 for naive masking: naive masking leaves the answer probability of non-final bridge hops unchanged (Δ=0.004), whereas graph-aware masking lowers it by 0.64, exactly as predicted. Placebo rewrites and a mask-marker control produce no effect (AUC 0.50), ruling out artifacts. On CLUTRR the same model is *causally decoupled* from its chain: even masking the final gold hop leaves its answer distribution unchanged. We then turn the measurement into a training-free decoding rule (HGCD) that contrasts the model against interventions on its own hops, and we characterise when such contrastive decoding can help at all. At 1.5B it does not: HGCD matches chain-of-thought on both datasets, and the predicted robustness advantage over context-aware decoding under distractor injection is *not* observed. We report these negative results with their causal explanation: readout-level decoding can only amplify a dependence that exists, and cannot repair chains that fail upstream. All interventions are binary and programmatic, all code runs on a free Kaggle GPU, and the hypotheses for each round were fixed before the GPU runs.

1 Introduction

Chain-of-thought prompting [33] makes models write intermediate steps, but writing a step is not the same as *using* it. Causal tests show that models frequently ignore or post-rationalise their own chains [9, 15, 28, 29], answer multi-hop questions through single-hop shortcuts [2, 13, 35], and are easily derailed by irrelevant context [20]. Two families of remedies exist. *Training* remedies (counterfactual preference objectives [8, 15], PNS-filtered supervision [36], causality-aware post-training [6]) are effective but costly. *Decoding* remedies are training-free: contrastive decoding [11, 14], classifier-free guidance [19], and context-aware decoding (CAD) [21] with its adaptive and gated successors [25, 31].

CAD scores tokens by $(1{+}\alpha)\log \pi(\cdot \mid c, x) - \alpha \log \pi(\cdot \mid x)$. Read causally, the log-ratio is the effect of do($C{=}c$) versus do($C{=}\varnothing$): the total effect of the *entire* context. In multi-hop QA the context contains supporting paragraphs on the causal path to the answer and distractors that are not; CAD amplifies both. This offers a causal explanation for the empirical failures of CAD on uninformative or distracting context that motivated AdaCAD and NWCAD [25, 31].

Idea. Instead of contrasting *with versus without context*, contrast the model's answer distribution *given its own chain* against the distribution under *interventions on the hops of that chain*, keeping the context fixed. Because the context is held fixed in both terms, distractor and parametric components cancel; because interventions follow the chain's own graph, the contrast measures the *path-specific* contribution of the reasoning steps.

Contributions.

1. **A causal account of contrastive decoding** for multi-hop reasoning: CAD estimates a total context effect that includes non-causal distractor paths (proposition 3).
2. **HGCD**, a training-free rule that contrasts the model against graph-aware interventions on its own reasoning chain, with an optional evidence-level variant (L1) that uses a *type-matched counterfactual context* instead of an empty one (section 4).
3. **Theory**: naive step masking measures a controlled direct effect that is zero on bridge chains (proposition 1); probability-space contrast cancels shortcut and distractor mixture components exactly (proposition 2); HGCD converges to a hop-level PNS rule (proposition 4), connecting decoding to the identifiability results of Yu et al. [36].
4. **A verifiable evaluation protocol**: binary, programmatic interventions with known ground-truth effects, an estimator-validation benchmark with known necessity labels, placebo controls, and paired statistics; everything reproducible on one free T4 GPU.
5. **Empirical findings at 1.5B** (section 7): the graph-aware estimator is valid where the naive one fails (MuSiQue); the model is causally decoupled from its chain on CLUTRR; and readout-level contrastive decoding, including ours, does not improve accuracy or robustness, for reasons the causal analysis makes explicit.

2 Related Work

Faithfulness of reasoning chains. Lanham et al. [9] intervene on model-written chains (truncation, inserted mistakes, paraphrases); Turpin et al. [28] show biased features steer answers without being verbalised; FRODO [15] uses causal mediation [30] and trains small models with counterfactual preferences; FUR [29] measures faithfulness by unlearning steps; Thought Anchors [3] measures step importance by counterfactual resampling, and Thought Branches [12] argues that step effects must be measured by resampling the continuation (a total effect), noting that removing a sentence while preventing its content from reappearing changes conclusions. FRODO already treats the rationale as a mediator and finds weak indirect effects for models under 20B, and Somov et al. [23] find that models often fail to update after edits to intermediate structures. Our contribution relative to these is narrower and specific: (i) necessity labels and null effects that are *known by construction*, so step-importance estimators can be *validated* rather than compared with each other; (ii) a deterministic, rollout-free total-effect intervention for hop-structured chains, validated against resampling; (iii) the bridge-versus-compositional distinction, which predicts exactly where direct-effect estimators fail; and (iv) an exact mediation of distractor effects (chain vs. readout).

Necessity and sufficiency. Yu et al. [36] define step-level probabilities of necessity and sufficiency for chain-of-thought, prove identifiability under exogeneity and monotonicity (building on Pearl [17], Tian and Pearl [26]), and use PNS to prune chains for in-context learning and fine-tuning. We show the same quantity arises as the limit of a decoding rule, and we separate direct from total effects [16].

Table 1: Positioning against closely related work. "GT" = the true value of the measured effect is known by construction.

Work	What is intervened on	Effect measured	GT?	Training	Controls
Lanham et al. [9]	model's CoT (truncate, add mistake, paraphrase)	answer change	no	none	paraphrase, filler tokens
Stolfo et al. [24] (ACL)	input factors of math problems	total / direct causal effect	partly	none	–
Paul et al. [15] (EMNLP-F)	rationale	mediation (indirect effect)	no	preference training	–
Yu et al. [36] (NeurIPS)	CoT step, replaced + rolled out	PNS	no (LLM judge)	SFT / ICL	–
Macar et al. [12] (ICLR)	sentence, resampled continuation	total effect, resilience	no	none	on- vs. off-policy
Somov et al. [23]	intermediate structure (rubric)	decision update rate	yes (deterministic map)	none	–
Tutek et al. [29] (EMNLP)	step information in weights	prediction change	no	unlearning	–
Shi et al. [21], Wang et al. [31]	whole context	(implicit) total context effect	no	none (decoding)	–
Ours	hop + DRG descendants; binary input interventions	direct vs. total effect; NDE/NIE	**yes**	none	placebo, mask-marker, redundant hop, content-free

Contrastive and guided decoding. Contrastive decoding [11, 14], DoLa [4], CFG [19], CAD and its adaptive/gated variants [21, 25, 31] contrast distributions under different models, layers or contexts. Visual contrastive decoding [10] uses a *distorted* rather than empty input; our L1 variant is the textual analogue with a type-matched counterfactual. Pced [5] treats each retrieved document as an expert contrasted with a no-context baseline; RED [34] multiplies image- and rationale-conditional distributions in vision-language models. None of these intervene on the hops of the model's own reasoning graph, distinguish direct from total effects, or relate the contrast to PNS.

Causal analyses of reasoning. Stolfo et al. [24] quantify robustness of mathematical reasoning through interventions on a causal graph of the input; Zečević et al. [37] argue models "talk" causality without being causal; text-intervention effect estimation [7] uses LLM rewrites as treatments. Our interventions are programmatic, so their ground-truth effects are known exactly.

3 Setup: a Structural Causal Model of Multi-Hop Reasoning

Variables. A question Q, a context $P = (P_{\text{sup}}, P_{\text{dist}})$ of supporting and distractor passages, a chain of hops $H_1, \ldots, H_k$, and an answer A. Each hop is rendered as `Hop` i`:` $\langle$`body`$\rangle$ `=>` $\langle$`result`$\rangle$. The model acts as a *reader* mechanism $\pi(a \mid Q, P, h_{1:k})$ over a finite candidate set $\mathcal{A}$.

Two task graphs. In *bridge* chains (MuSiQue), hop j resolves an entity that the next hop consumes and the answer is the result of the last hop: $Q \to H_1 \to \cdots \to H_k \to A$, with $P_{\text{sup},j} \to H_j$. In *compositional* chains (CLUTRR), every hop contributes a relation that is composed into the

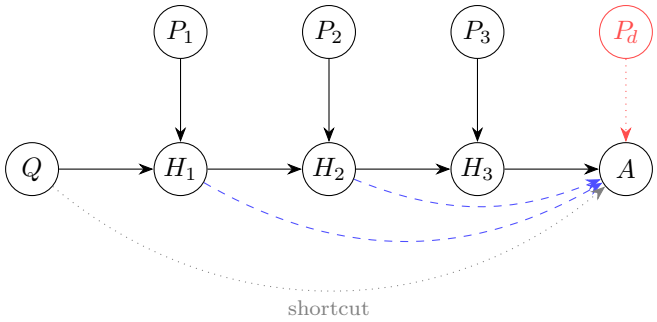


Figure 1: Task SCM. Solid: bridge chain. Dashed blue: extra edges of compositional chains. Red dotted: spurious distractor path that the *reader* may use but the task graph forbids. Gray: parametric shortcut. CAD's contrast includes the red path; HGCD holds P fixed and contrasts interventions on H_j only.

answer: $H_j \to A$ for all j and $H_j \to H_{j+1}$ through the shared entity. Distractors have no edge to A in either task graph (fig. 1).

Directed reasoning graph (DRG). Given a generated chain $y = (h_1, \ldots, h_k)$ we add an edge $i \to j$ $(i < j)$ if the result of h_i is mentioned in the body of h_j. Terminal hops are those whose result is the query target (CLUTRR) or the last hop (MuSiQue); the *relevant set* $\mathcal{R}(y)$ contains the terminal hops and their DRG ancestors. Hops outside $\mathcal{R}(y)$ have no directed path to the answer.

Interventions on hops.

Definition 1 (Naive and graph-aware interventions)**.** y_j^{nai} *replaces hop* j *by a mask token and leaves all other hops unchanged.* y_j^{tot} *masks hop* j *and, in every DRG descendant* d *of* j*, masks every mention of the results of* j *and of* d*. The latter is a deterministic, rollout-free approximation of* $\text{do}(H_j{=}\perp)$*: descendant values are functions of* h_j*, hence unknown under the intervention. A* placebo *rewrites hop* j *without changing its entities or relation.*

For an answer a we write $\Delta_j^{\text{nai}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{nai}})$ and $\Delta_j^{\text{tot}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{tot}})$, suppressing Q, P.

What is (and is not) causal here. We do not assume the language model *is* an SCM of the task. The reader π is a deterministic function of its input, so editing hops in the prompt is an exact intervention on the reader's input, and Δ_j is an exact (not estimated) effect *on the reader*. The task SCM only supplies ground truth: which hops are necessary, and which input changes should have no effect on the answer.

Mask-token artifacts. Replacing text by `[MASKED]` is itself out of distribution. Our estimator benchmark controls for this: the inserted redundant hop is masked in exactly the same way, so any effect caused by the mask token alone appears as a non-zero Δ on redundant hops and lowers the AUC. In addition, a *mask-marker* control inserts `[MASKED]` into hop j while keeping all of its content; its effect Δ_j^{marker} measures the mask-token artifact directly and should be ≈ 0.

Algorithm 1 HGCD (L2) for one question

$y \leftarrow$ greedy chain from the model (optionally with L1 contrast on result tokens)
Parse hops; build DRG; compute relevant set $\mathcal{R}(y)$
$p \leftarrow \pi(\cdot \mid Q, P, y)$ over $\mathcal{A}$ ▷ one forward + KV-cached candidates
for $j \in \mathcal{R}(y)$ **do**
    $p_j \leftarrow \pi(\cdot \mid Q, P, y_j^{\text{tot}})$
end for
return $\arg\max_{a \in \mathcal{V}_\beta} (1+\alpha) p(a) - \alpha \operatorname{mean}_j p_j(a)$

4 Method: Hop-Graph Contrastive Decoding

Answer-level rule (L2). With uniform weights $w_j = 1/|\mathcal{R}(y)|$ on relevant hops and a plausibility set $\mathcal{V}_\beta = \{a : \pi(a \mid y) \geq \beta \max_{a'} \pi(a' \mid y)\}$ [11],

$$\hat{a} = \arg\max_{a \in \mathcal{V}_\beta} s_\alpha(a),$$
$$s_\alpha(a) = (1+\alpha)\,\pi(a \mid y) - \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \pi(a \mid y_j^{\text{tot}}) = \pi(a \mid y) + \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \Delta_j^{\text{tot}}(a). \tag{1}$$

The score is the answer probability plus α times the average total effect of the answer-relevant hops on that answer. Redundant hops ($j \notin \mathcal{R}(y)$) are never contrasted.

Evidence-level rule (L1). While emitting the *result* tokens of a hop, we contrast the real context P with a type-matched counterfactual $\tilde{P}$ of identical length and entity types in which the relational content differs (on CLUTRR: names permuted within each gender group, so every sentence and entity type is kept but who-is-related-to-whom changes; we evaluate L1 on CLUTRR only, where $\tilde{P}$ is exactly type-matched): $t^\star = \arg\max_{t \in \mathcal{V}_\beta} (1+\alpha)\pi(t \mid P, \cdot) - \alpha\pi(t \mid \tilde{P}, \cdot)$. Unlike an empty context, $\tilde{P}$ is in-distribution and controls for length and type cues.

Remark 1 (What each level can fix)**.** *L2 makes the answer depend* more *on the chain; it therefore repairs* readout *errors (chain correct, answer wrong because a shortcut or distractor dominates the readout) but cannot repair a wrong chain—it may even commit harder to one. Chain errors are the target of L1, which acts while the hops are produced. We test this split directly (hypothesis 4).*

Cost. L2 needs $1 + |\mathcal{R}(y)|$ prefix forwards and a KV-cached pass over the candidate tokens; no extra generation. L1 doubles the forward cost of result tokens only.

Choice of α. α is selected on the CLUTRR validation split and frozen (ties broken toward the smaller α, so HGCD reduces to CoT absent evidence); an adaptive variant sets $\alpha \cdot \mathrm{JSD}(p, \bar{p}_j)$ as in AdaCAD [31].

5 Theory

All proofs are short and given in appendix A; the propositions make predictions that our protocol tests directly.

Assumption 1 (Markov reader on bridge chains)**.** *For bridge chains, $\pi(a \mid h_{1:k})$ depends on $h_{1:j}$ only through $h_{j+1:k}$ whenever h_{j+1} restates the result of h_j ($j < k$).*

Proposition 1 (Naive masking is a direct effect)**.** $\Delta_j^{\text{nai}}(a)$ *equals the controlled direct effect of* H_j *on* A *with* $H_{j+1:k}$ *held at their observed values. Under assumption 1,* $\Delta_j^{\text{nai}}(a) = 0$ *for every* $j < k$ *on bridge chains, whereas* $\Delta_j^{\text{tot}}(a)$ *is not constrained to zero. On compositional chains, where* A *reads each hop's relation, the direct effect is generally non-zero.*

Prediction. On CLUTRR both estimators separate necessary from redundant hops; on MuSiQue, $\Delta^{\text{nai}} \approx 0$ on non-final gold hops while Δ^{tot} remains large, so the gap between the two appears in the position breakdown of table 3.

Assumption 2 (Mixture reader)**.** $\pi(a \mid Q, P, y) = \lambda_c \pi_c(a \mid y, P_{\text{sup}}) + \lambda_s \pi_s(a \mid Q) + \lambda_d \pi_d(a \mid P_{\text{dist}})$*, with* λ*'s non-negative, summing to one, and invariant to interventions on hops.*

Proposition 2 (Exact cancellation in probability space)**.** *Under assumption 2,*

$$s_\alpha(a) = \pi(a \mid y) + \alpha\lambda_c \sum_j w_j \big[\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})\big],$$

i.e. the shortcut π_s *and distractor* π_d *components cancel in the contrast for every* α*. The log-space analogue does not cancel them.*

Proposition 3 (CAD amplifies distractor paths)**.** *Under assumption 2 applied to direct answering, with no-context distribution* $\pi_s(a \mid Q)$*, CAD's score is* $(1{+}\alpha)\log\big[\lambda_c\pi_c{+}\lambda_s\pi_s{+}\lambda_d\pi_d\big](a){-}\alpha\log\pi_s(a \mid Q)$*. For any* $\alpha > 0$ *and any gold answer* g*, a distractor answer* a_d *with* $\pi_d(a_d) = 1$ *is preferred by CAD whenever* $\pi_s(a_d \mid Q)$ *is small enough, even if* $\pi(g \mid Q, P) > \pi(a_d \mid Q, P)$*. In contrast, HGCD's contrast term does not depend on* π_d *at all (proposition 2).*

Prediction. Under distractor injection, CAD's accuracy degrades more than HGCD's, i.e. $|\text{ATE}_{\text{distract}}|$ is larger for CAD.

Proposition 4 (PNS limit)**.** *Interpret* $\pi(a \mid y_j^{\text{tot}})$ *as* $P(A{=}a \mid \text{do}(H_j{=}\perp))$ *and assume exogeneity and monotonicity as in Yu et al. [36]. Then* $\Delta_j^{\text{tot}}(a) = \text{PNS}_j(a)$ *and* $\lim_{\alpha\to\infty} s_\alpha(a)/\alpha = \sum_{j\in\mathcal{R}(y)} w_j \,\text{PNS}_j(a)$*. Hence, within* $\mathcal{V}_\beta$*, HGCD interpolates between the most probable answer (*$\alpha{=}0$*) and the answer for which the relevant hops are, on average, most necessary and sufficient.*

6 Experimental Protocol

Data. CLUTRR [22] (`gen_train234_test2to10`), restricted to clean simple-path stories (the filter is reported), gives exact gold paths; ID $k \in \{2, 3, 4\}$, OOD $k \geq 5$; answers are scored over the 20 kinship relations. **MuSiQue-Ans** [27] gives real text with gold decompositions; the context holds the supporting paragraphs plus 4 distractors. MuSiQue was built to resist single-hop shortcuts [13], which is why we prefer it to HotpotQA. Candidates for MuSiQue are built without the gold answer: the model's own answer and hop results, then paragraph titles, capitalised spans and dates/numbers from the context, in order of appearance, capped at 64 (recall of the gold answer is 56% on 600 dev questions before adding the model's strings, versus 10.5% for titles alone). All methods share the same candidate set and we report its recall.

Binary interventions with known effects.

Intervention	Definition (programmatic, seeded)	True effect on A
`rename`	all names → fresh fictional names (gender-preserving)	0
`order`	shuffle story sentences / paragraphs	0
`distract` / `dist8`	insert a disconnected family / 4 extra distractor paragraphs	0
`placebo_input`	insert two neutral sentences without entities	0
redundant hop	insert a true but off-path fact as a hop of a gold chain	necessity 0

For each method we report the paired ATE $\mathbb{E}[Y(1) - Y(0)]$ on accuracy with bootstrap 95% CIs; a robust method has ATE ≈ 0. `placebo_input` controls for "any change hurts".

Estimator validation. Gold chains with one inserted redundant hop give hop-level labels (necessary = 1, redundant = 0). We report AUC of Δ^{tot}, Δ^{nai}, the placebo and mask-marker Δ, rollout PNS [36] and the DRG path indicator, overall and on chains the model answers correctly (PS = 1), and mean effects by hop position.

Methods. Direct answering; CoT; self-consistency [32] with 5 sampled chains (strictly more compute than HGCD, which generates nothing extra); CAD [21]; AdaCAD [31]; CAD-chain (contrast with vs. without the chain, a text analogue of rationale-conditioned decoding); HGCD and ablations: naive masking, no graph (all hops), log space, adaptive α, placebo contrast (the key control); L1 chains decoded with CoT or HGCD.

Hypotheses. The following hypotheses and decision criteria were fixed in the repository before any GPU experiment was run (verifiable in the commit history).

Hypothesis 1 (Estimator)**.** *AUC*$(\Delta^{\text{tot}}) \geq 0.75$ *on both datasets;* Δ^{nai} *near zero on MuSiQue non-final gold hops; placebo and mask-marker* $\Delta \approx 0$ *on all hops.*

Hypothesis 2 (Accuracy)**.** *HGCD > CoT and > CAD-chain on OOD CLUTRR (*$k \geq 5$*) and on MuSiQue, with paired bootstrap CIs excluding zero.*

Hypothesis 3 (Robustness)**.** $|\text{ATE}_{\text{distract}}|$ *of HGCD < that of CAD (proposition 3); ATE of* `placebo_input` ≈ 0 *for all methods.*

Hypothesis 4 (Readout)**.** *The gain of HGCD over CoT is concentrated on questions whose greedy chain reaches every gold-path entity, and is* ≈ 0 *(or negative) on the others (remark 1).*

Hypothesis 5 (Ablations)**.** *HGCD* $\geq$ *naive-mask and no-graph variants; probability space* $\geq$ *log space; and, crucially, HGCD > placebo contrast. Contrasting against* any *perturbation can move decisions (our CPU pilot shows this), so the placebo contrast, not CoT, is the control that isolates the causal content of the intervention.*

Models and compute. Qwen2.5-1.5B-Instruct [18] (main) and SmolLM2-1.7B-Instruct [1] (second family) on two Kaggle T4 GPUs in fp16; Qwen2.5-7B-Instruct on one A100 for the scale point. Greedy decoding; $\beta = 0.1$; $\alpha \in \{0, 0.25, 0.5, 1, 2, 4\}$ chosen per method on the CLUTRR validation split and transferred unchanged to MuSiQue (no MuSiQue data is used for tuning).

Statistics. Bootstrap CIs (2,000 resamples), paired bootstrap differences and exact McNemar tests on the same questions. The three *primary* comparisons are HGCD vs. CoT, vs. CAD-chain and vs. the placebo contrast, on OOD CLUTRR and on MuSiQue; all other contrasts are exploratory and are reported without multiplicity claims. With n questions per split, the half-width of a 95% CI on accuracy is about $1/\sqrt{n}$; paired tests are considerably tighter.

7 Results

All tables in this section are generated by the notebook; nothing is typed by hand. Model: Qwen2.5-1.5B-Instruct, fp16, one Kaggle T4; 238 CLUTRR test questions (118 ID, 120 OOD) and 120 MuSiQue questions; 90 CLUTRR and 80 MuSiQue gold chains for estimator validation (360 and 297 hops). Run time: 195 minutes.

Table 2: Hypothesis scorecard for round 1 (Qwen2.5-1.5B). ✓: supported; ×: not supported; –: not evaluable from this run.

Hypothesis	Evidence	Verdict
H1 Estimator (MuSiQue)	AUC(Δ^{tot}) = 0.971 vs. 0.789 naive; non-final gold hops: $\Delta^{\text{tot}} = 0.639$, $\Delta^{\text{nai}} = 0.004$; placebo and marker AUC 0.50, $\lvert\Delta\rvert \leq 0.003$	✓
H1 Estimator (CLUTRR)	all AUCs 0.47–0.54; $\lvert\Delta\rvert \leq 0.017$ even for the final gold hop; model correct with the gold chain on only 67/360 hops	× (reader decoupled)
H2 Accuracy	CLUTRR: HGCD = CoT (18.1%, selected α=0.25 leaves decisions unchanged); MuSiQue: 22.5 vs. 20.8 EM, CIs overlap	×
H3 Robustness	MuSiQue `dist8`: CAD ATE −0.8, HGCD −7.5, CoT −9.2 [−16.7,−2.5]; CLUTRR: all ATE CIs contain 0; `placebo_input` ATE ≈ 0 for every method	×
H4 Readout	table not printed in this run's log	–
H5 Ablations	HGCD < naive-mask (−5.0, p=0.004) and < log-space (−4.6, p=0.04) on CLUTRR, both at the dev-selected α=4, in a regime where H1 fails	×

The estimator is valid where the reader uses its chain (H1, MuSiQue). Graph-aware masking identifies necessary hops almost perfectly, and the position breakdown (table 3) reproduces the qualitative prediction of proposition 1 with 297 hops: naive masking measures a direct effect that is essentially zero on non-final bridge hops (0.004), while the total effect is 0.639, close to that of the final hop (0.679). The placebo and mask-marker controls are flat, so the effects are not produced by the `[MASKED]` token or by rewording.

On CLUTRR the reader is causally decoupled from its chain (H1, CLUTRR). Masking *any* hop of a gold chain, including the final one, changes the probability of the gold relation by less than 0.02 on average, and the model answers correctly from the gold chain for only about 19% of chains (67 of 360 hop rows). The estimator cannot separate necessary from redundant hops because *no* hop has an effect: the chain is decorative for this reader. This is a finding about the model, not a failure of the estimator (the DRG indicator, which uses only the chain's structure, has AUC 1.0). Accuracy of every method (15.5–23.1%) is close to always predicting the most frequent relation in this sample (15.1%), and model-written chains form a connected DRG in only 25% of cases.

Contrastive decoding cannot amplify a dependence that does not exist (H2, H5). On CLUTRR the dev-selected α for HGCD is 0.25, which leaves every decision unchanged. The naive-mask and log-space variants, selected at α=4, gain 4–5 points over CoT; because their contrast

terms compare two nearly identical distributions (H1 fails here), we treat these gains as a re-weighting of the label prior rather than causal amplification, and test this explicitly in round 2 with a content-free calibration baseline (H8).

Distractors act through the chain, not the readout (H3). On MuSiQue, adding four distractor paragraphs lowers CoT's exact match by 9.2 points, while CAD, which answers without a chain, is unaffected (-0.8). proposition 3 concerns the readout with the chain held fixed, but this ATE also lets the chain be regenerated under distractors. The pattern is what one expects if distractors mainly corrupt *chain generation*, which no readout-level rule can undo (remark 1). Round 2 separates the two paths with an exact mediation analysis (H6).

7.1 What round 1 teaches: four insights

I1. Step-masking faithfulness tests are blind to bridge reasoning. On MuSiQue, 99.4% of the causal effect of a non-final hop is indirect ($\Delta^{\text{nai}} = 0.004$ vs. $\Delta^{\text{tot}} = 0.639$), and the total effect of a non-final hop is 94% of that of the final hop. Any metric that ablates one step while keeping the rest of the chain (ablation accuracy, step perplexity or information gain computed on a fixed continuation) would declare every intermediate bridge step *unnecessary* although each is almost as necessary as the last. The bias is not noise: it is predicted in closed form by proposition 1 and appears exactly where the theory says it should (bridge, not compositional, graphs).

I2. Small models copy from their chain; they do not compute over it. The two datasets differ in one structural respect: in MuSiQue the answer is a *string written in the last hop*; in CLUTRR the answer is a relation that must be *composed* and never appears in the chain. The same 1.5B reader depends strongly on the chain in the first case (final-hop effect 0.68) and not at all in the second (≈ 0, even with a gold chain). This suggests that reliance on a chain is a property of the *answer's form* rather than of the task's difficulty. A binary intervention isolates it: appending one hop that writes the answer into an otherwise identical CLUTRR chain (T_{copy}=1). In a CPU sanity check with a 0.5B model on four questions (an illustration, not evidence), the content effect of the chain jumps from 0.003 to 0.59 and CLUTRR switches to the bridge pattern of I1 (non-final hops: $\Delta^{\text{tot}} = 0.71$, $\Delta^{\text{nai}} = 0.06$). Round 2 measures this at scale (H11).

I3. Faithfulness amplifies robustness failures. Under distractor injection on MuSiQue, methods that answer *through* a chain lose on average 7.1 EM points (CoT -9.2), while methods that answer without one lose 1.6 (CAD -0.8). At this scale chain-free direct answering is also more accurate than CoT (25.0 vs. 20.8 EM). When the reader copies from its chain (I2), every error the generator makes under distraction is propagated faithfully. Faithful reasoning is therefore only as good as the chain generator, and "making models use their chains" is not unconditionally desirable. The mediation analysis (H6) tests whether this loss is indeed indirect.

I4. A decoding rule can only amplify dependence that exists. HGCD adds α times the total effect of the chain to each answer's score. On CLUTRR this effect is ≈ 0 (nothing to amplify); on MuSiQue the reader already copies the final hop (effect saturated). Both regimes predict a null gain, which is what we observe. The account turns a negative result into a falsifiable prediction: gains require *intermediate* reliance (H-niche), which can be created by scale or by the answer's form (I2).

7.2 Round-2 hypotheses (fixed before the next run)

Each hypothesis below is computed by the updated notebook; the mediator (the chain) is set exactly by feeding the original chain under the intervened input, so the natural direct effect NDE = $\mathbb{E}[Y(1, M(0)) - Y(0, M(0))]$ is observed per question without cross-world assumptions, and NIE = TE − NDE [16].

Hypothesis 6 (Mediation)**.** *Under* `dist8`*, most of CoT's total effect is indirect:* |NIE| > |NDE|*; with the chain fixed, HGCD's NDE is no larger in magnitude than CAD-chain's (the proper test of proposition 3).*

Hypothesis 7 (Reliance grows with scale)**.** *Mean* Δ^{tot} *of gold hops on CLUTRR increases from 0.5B to 7B (Qwen2.5 0.5/1.5/3/7B), and AUC*(Δ^{tot}) *on CLUTRR exceeds* 0.75 *at 7B.*

Hypothesis 8 (Calibration)**.** *The* α=4 *gains of naive/log variants on CLUTRR are matched by content-free calibration [38] (contrast against the no-story, no-chain distribution).*

Hypothesis 9 (Estimator validity)**.** *On bridge chains, total-effect estimators (mask+descendants, counterfactual replacement with propagation, rollout PNS) all reach AUC* ≥ 0.9 *and agree with each other, while direct-effect estimators (mask hop only, replacement without propagation) stay near* 0 *on non-final hops; on compositional chains direct and total estimators coincide. The deterministic mask+descendants estimator matches rollout PNS at a fraction* ($\approx 1/K$) *of its cost.*

Hypothesis 10 (Copy, not compute)**.** *On CLUTRR gold chains, appending a hop that writes the answer (*T_{copy}=1*) raises the content effect of the chain from* ≈ 0 *to* > 0.3 *at every scale up to 3B, and makes* $\Delta^{nai} \approx 0$ *on the relational hops while* Δ^{tot} *stays large (the bridge pattern).*

Hypothesis 11 (Form vs. content)**.** *Where CoT beats direct answering although hop content has no effect (CLUTRR OOD:* 19.2 *vs.* 11.7*), the gain comes from the chain's form: a chain of the same shape with every hop masked shifts* $p(gold)$ *as much as the real chain.*

Hypothesis 12 (Faithfulness amplifies failures)**.** *Across methods, the distractor ATE is larger for chain-based than for chain-free readers, and for chain-based readers it is mostly indirect (*|NIE| > |NDE|*). A product of the direct and chain-based readers (PoE) is more robust than CoT at equal accuracy.*

Hypothesis 13 (Where decoding can help)**.** *The gain of HGCD over CoT is positive only for models with intermediate reliance (gold-hop* Δ^{tot} *neither* ≈ 0 *nor* $\approx$ *saturated).*

8 Limitations

Assumptions. ?? 1?? 2 idealise the reader; we treat the propositions as testable predictions rather than guarantees, and the protocol is designed to falsify them. **Rollout-free interventions.** y_j^{tot} approximates do(H_j) by masking rather than re-generating descendants; we validate it against rollout PNS where compute allows. **Format dependence.** The DRG requires hop-structured chains linked by explicit mentions; pronouns break links and shrink $\mathcal{R}(y)$ to the terminal hop. We report format success, hop fidelity against the gold path and the fraction of fully connected DRGs, and fall back to CoT when parsing fails. **Closed-set scoring.** All CLUTRR methods, including CoT, are scored over the same 20 relations, so differences are not due to answer extraction. **Candidate sets.** Answer-level contrast needs a finite candidate set; it is exact on CLUTRR and approximate on MuSiQue (we report candidate recall). **Scale.** Main experiments use 1.5–7B models; larger models may rely less on shortcuts. **Data filter.** We use CLUTRR stories whose gold graph is a simple path; stories with back-edges contain known label ambiguities.

Table 3: Estimator validation (AUC for separating necessary from inserted redundant hops).

Estimator	clutrr		musique	
	all	PS=1	all	PS=1
Δ^{naive} (mask hop only)	0.532	0.583	0.789	0.798
Δ^{total} (graph-aware mask)	0.523	0.539	0.971	0.972
DRG path indicator	1.000	1.000	0.994	0.994
Δ^{placebo} (content-preserving)	0.468	0.469	0.497	0.491
Δ^{marker} (mask token, content kept)	0.542	0.558	0.497	0.499

Dataset	Hop type	Δ^{total}	Δ^{naive}	Δ^{placebo}	Δ^{marker}
clutrr	gold-final	0.004	0.004	0.006	0.015
clutrr	gold-nonfinal	-0.017	0.000	-0.002	-0.010
clutrr	redundant	-0.005	-0.005	0.009	-0.006
musique	gold-final	0.679	0.679	-0.002	0.002
musique	gold-nonfinal	0.639	0.004	0.003	0.002
musique	redundant	0.007	-0.006	0.002	0.002

Table 4: CLUTRR accuracy (%) with bootstrap 95% CIs. ID: $k \leq 4$; OOD: $k \geq 5$.

Method	ID (n=118)	OOD (n=120)	All (n=238)
Direct (no CoT)	20.3 [13.6,28.0]	11.7 [5.8,17.5]	16.0 [11.3,20.6]
CoT	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
Self-consistency	21.2 [14.4,28.8]	17.5 [10.8,25.0]	19.3 [14.7,24.4]
CAD	20.3 [13.6,28.0]	17.5 [10.8,24.2]	18.9 [13.9,24.0]
AdaCAD	20.3 [13.6,27.1]	10.8 [5.0,16.7]	15.5 [10.9,20.2]
CAD-chain	18.6 [11.9,25.4]	17.5 [10.8,24.2]	18.1 [13.4,23.1]
HGCD (naive mask)	20.3 [13.6,28.0]	25.8 [18.3,33.3]	23.1 [17.6,28.6]
HGCD (no graph)	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
HGCD (log-space)	17.8 [11.0,24.6]	27.5 [19.2,35.0]	22.7 [17.6,28.2]
HGCD (adaptive α)	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
HGCD (placebo)	18.6 [11.9,25.4]	21.7 [15.0,29.2]	20.2 [15.5,25.2]
HGCD	16.9 [11.0,23.7]	19.2 [12.5,26.7]	18.1 [13.9,23.5]
L1 + CoT	19.5 [12.7,26.3]	17.5 [10.8,25.0]	18.5 [13.9,23.5]
L1 + HGCD	19.5 [12.7,26.3]	17.5 [10.8,25.0]	18.5 [13.9,23.5]

9 Conclusion

Measuring whether a model uses its reasoning steps requires interventions that respect the chain's graph: naive step masking measures a direct effect and misses intermediate bridge hops entirely. With the graph-aware estimator, we find that a 1.5B model relies heavily on its chain for real multi-hop QA but not at all on CLUTRR. Turning the measurement into a contrastive decoding rule does not improve accuracy at this scale, and our causal analysis says why: decoding at the readout can amplify only dependence that exists, and distractors harm reasoning mainly through the chain. The round-2 hypotheses test these explanations directly.

Table 5: CLUTRR paired ATE (accuracy points) of binary interventions whose true effect is 0.

Method	distract	order	placebo_input	rename
Direct (no CoT)	-2.5 [-6.3,0.8]	0.0 [-4.2,4.2]	0.8 [-0.8,2.5]	2.1 [-0.8,5.5]
CoT	-1.3 [-6.3,3.8]	0.4 [-5.0,5.5]	-0.4 [-5.0,3.8]	2.5 [-2.9,8.0]
CAD	0.4 [-4.2,5.0]	1.7 [-3.4,6.7]	0.8 [-2.5,4.2]	0.8 [-4.2,5.5]
AdaCAD	-0.4 [-4.2,3.4]	0.4 [-4.2,4.6]	0.4 [-1.3,2.1]	2.5 [-0.8,6.3]
CAD-chain	-2.5 [-8.0,2.9]	2.5 [-3.4,8.4]	2.1 [-2.9,6.7]	2.9 [-2.5,8.4]
HGCD (no graph)	-1.7 [-6.7,3.4]	1.7 [-4.2,7.1]	0.0 [-4.6,4.2]	2.5 [-2.9,8.0]
HGCD (log-space)	-3.4 [-10.5,2.9]	-1.7 [-8.0,5.0]	-0.4 [-5.9,5.0]	1.7 [-4.6,8.0]
HGCD (adaptive α)	-1.3 [-6.3,3.8]	1.3 [-4.2,6.7]	0.0 [-4.6,4.2]	2.5 [-2.9,8.0]
HGCD	-1.7 [-6.7,3.4]	1.3 [-4.2,6.7]	1.3 [-3.4,5.5]	2.5 [-2.9,8.0]

Table 6: CLUTRR accuracy (%) split by whether the greedy chain reaches every gold-path entity (hypothesis 4).

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

References

[1] Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, et al. SmolLM2: When smol goes big – data-centric training of a small language model. *arXiv preprint arXiv:2502.02737*, 2025.

[2] Eden Biran, Daniela Gottesman, Sohee Yang, Mor Geva, and Amir Globerson. Hopping too late: Exploring the limitations of large language models on multi-hop queries. In *Proceedings of EMNLP*, pages 14113–14130, 2024.

[3] Paul C. Bogdan, Uzay Macar, Neel Nanda, and Arthur Conmy. Thought anchors: Which LLM reasoning steps matter? *arXiv preprint arXiv:2506.19143*, 2025.

[4] Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. DoLa: Decoding by contrasting layers improves factuality in large language models. In *Proceedings of ICLR*, 2024.

[5] Giulio Corallo and Paolo Papotti. Parallel context-of-experts decoding for retrieval augmented generation. *arXiv preprint arXiv:2601.08670*, 2026.

[6] Shurui Gui and Shuiwang Ji. Mitigating spurious correlations in LLMs via causality-aware post-training. *arXiv preprint arXiv:2506.09433*, 2025.

[7] Siyi Guo, Myrl G. Marmarelis, Fred Morstatter, and Kristina Lerman. Estimating causal effects of text interventions leveraging LLMs. *arXiv preprint arXiv:2410.21474*, 2024.

[8] Alihan Hüyük, Xinnuo Xu, Jacqueline Maasch, Aditya V. Nori, and Javier González. Reasoning elicitation in language models via counterfactual feedback. In *Proceedings of ICLR*, 2025.

[9] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.

Table 7: MuSiQue exact match (%) and paired ATE.

Method	All (n=120)
Direct (no CoT)	25.0 [17.5,33.3]
CoT	20.8 [14.2,28.3]
CAD	24.2 [16.7,31.7]
AdaCAD	25.0 [17.5,33.3]
CAD-chain	21.7 [15.0,29.2]
HGCD (naive mask)	24.2 [17.5,31.7]
HGCD (no graph)	21.7 [15.0,29.2]
HGCD (log-space)	23.3 [16.7,30.8]
HGCD (adaptive α)	21.7 [15.0,29.2]
HGCD (placebo)	21.7 [15.0,29.2]
HGCD	22.5 [15.8,30.0]
Free generation	23.3 [16.7,30.8]

Method	dist8	order
Direct (no CoT)	-3.3 [-10.0,2.5]	-4.2 [-10.8,2.5]
CoT	-9.2 [-16.7,-2.5]	0.0 [-6.7,6.7]
CAD	-0.8 [-7.5,5.8]	-4.2 [-11.7,3.3]
AdaCAD	-0.8 [-8.3,5.8]	-4.2 [-11.7,3.3]
CAD-chain	-5.8 [-13.3,0.8]	0.8 [-5.8,8.3]
HGCD (no graph)	-6.7 [-14.2,0.8]	4.2 [-3.3,11.7]
HGCD (log-space)	-7.5 [-15.0,-0.8]	0.0 [-7.5,6.7]
HGCD (adaptive α)	-7.5 [-15.8,0.0]	2.5 [-5.8,10.8]
HGCD	-7.5 [-15.0,0.0]	2.5 [-5.0,10.0]
Free generation	-5.8 [-14.2,1.7]	0.0 [-7.5,7.5]

Table 8: Round 2: reliance and estimator validity across model scales (estimator benchmark). PS: fraction of gold chains answered correctly; Rel.: mean total effect of gold hops on p(gold); remaining columns: AUC of each estimator for necessary vs. redundant hops.

Pending: run `notebooks/hgcd_kaggle.ipynb` and copy its output folder to `results/kaggle_main`.

[10] Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *Proceedings of CVPR*, 2024.

[11] Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive decoding: Open-ended text generation as optimization. In *Proceedings of ACL*, 2023.

[12] Uzay Macar, Paul C. Bogdan, Senthooran Rajamanoharan, and Neel Nanda. Thought branches: Interpreting LLM reasoning requires resampling. In *Proceedings of ICLR*, 2026.

[13] Sewon Min, Eric Wallace, Sameer Singh, Matt Gardner, Hannaneh Hajishirzi, and Luke Zettlemoyer. Compositional questions do not necessitate multi-hop reasoning. In *Proceedings of ACL*, 2019.

[14] Sean O'Brien and Mike Lewis. Contrastive decoding improves reasoning in large language models. *arXiv preprint arXiv:2309.09117*, 2023.

[15] Debjit Paul, Robert West, Antoine Bosselut, and Boi Faltings. Making reasoning matter: Measuring and improving faithfulness of chain-of-thought reasoning. In *Findings of EMNLP*, pages 15012–15032, 2024.

[16] Judea Pearl. Direct and indirect effects. In *Proceedings of UAI*, pages 411–420, 2001.

[17] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2 edition, 2009.

[18] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

[19] Guillaume Sanchez, Alexander Spangher, Honglu Fan, Elad Levi, and Stella Biderman. Stay on topic with classifier-free guidance. In *Proceedings of ICML*, 2024.

[20] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context. In *Proceedings of ICML*, 2023.

[21] Weijia Shi, Xiaochuang Han, Mike Lewis, Yulia Tsvetkov, Luke Zettlemoyer, and Wen-tau Yih. Trusting your evidence: Hallucinate less with context-aware decoding. In *Proceedings of NAACL (Short Papers)*, pages 783–791, 2024.

[22] Koustuv Sinha, Shagun Sodhani, Jin Dong, Joelle Pineau, and William L. Hamilton. CLUTRR: A diagnostic benchmark for inductive reasoning from text. In *Proceedings of EMNLP-IJCNLP*, 2019.

[23] Oleg Somov, Mikhail Chaichuk, Mikhail Seleznyov, Alexander Panchenko, and Elena Tutubalina. Breaking the chain: A causal analysis of LLM faithfulness to intermediate structures. *arXiv preprint arXiv:2603.16475*, 2026.

[24] Alessandro Stolfo, Zhijing Jin, Kumar Shridhar, Bernhard Schölkopf, and Mrinmaya Sachan. A causal framework to quantify the robustness of mathematical reasoning with language models. In *Proceedings of ACL*, pages 545–561, 2023.

[25] Yufei Tao and Ameeta Agrawal. No-worse context-aware decoding: Preventing neutral regression in context-conditioned generation. In *Findings of ACL*, 2026.

[26] Jin Tian and Judea Pearl. Probabilities of causation: Bounds and identification. *Annals of Mathematics and Artificial Intelligence*, 28:287–313, 2000.

[27] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022.

[28] Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Proceedings of NeurIPS*, 2023.

[29] Martin Tutek, Fateme Hashemi Chaleshtori, Ana Marasović, and Yonatan Belinkov. Measuring chain of thought faithfulness by unlearning reasoning steps. In *Proceedings of EMNLP*, 2025.

[30] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Proceedings of NeurIPS*, 2020.

[31] Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. AdaCAD: Adaptively decoding to balance conflicts between contextual and parametric knowledge. In *Proceedings of NAACL*, 2025.

[32] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *Proceedings of ICLR*, 2023.

[33] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Proceedings of NeurIPS*, 2022.

[34] Shin'ya Yamaguchi, Kosuke Nishida, and Daiki Chijiwa. Rationale-enhanced decoding for multi-modal chain-of-thought. In *Proceedings of CVPR*, 2026.

[35] Sohee Yang, Elena Gribovskaya, Nora Kassner, Mor Geva, and Sebastian Riedel. Do large language models latently perform multi-hop reasoning? In *Proceedings of ACL*, pages 10210–10229, 2024.

[36] Xiangning Yu, Zhuohan Wang, Linyi Yang, Haoxuan Li, Anjie Liu, Xiao Xue, Jun Wang, and Mengyue Yang. Causal sufficiency and necessity improves chain-of-thought reasoning. In *Proceedings of NeurIPS*, 2025.

[37] Matej Zečević, Moritz Willig, Devendra Singh Dhami, and Kristian Kersting. Causal parrots: Large language models may talk causality but are not causal. *Transactions on Machine Learning Research*, 2023.

[38] Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of ICML*, 2021.

A Proofs

Proposition 1. y_j^{nai} changes only h_j and keeps $h_{j+1:k}$ at their observed values, so Δ_j^{nai} is by definition the controlled direct effect $\pi(a \mid h_j, h_{-j}) - \pi(a \mid \perp, h_{-j})$ [16]. Under assumption 1, $\pi(a \mid h_{1:k}) = g(a \mid h_{j+1:k})$ for $j < k$, so both terms equal $g(a \mid h_{j+1:k})$ and the difference is zero. y_j^{tot} additionally changes $h_{j+1:k}$ (their results are masked), so g is evaluated at a different argument and the difference is unconstrained. On compositional chains π reads the relation of every hop, so removing h_j alone changes the composed relation. □

Proposition 2. Substituting assumption 2 into eq. (1), with $\sum_j w_j = 1$ and π_s, π_d unaffected by interventions on hops: $s_\alpha(a) = (1{+}\alpha)[\lambda_c \pi_c(a \mid y) + \lambda_s \pi_s + \lambda_d \pi_d] - \alpha \sum_j w_j [\lambda_c \pi_c(a \mid y_j^{\text{tot}}) + \lambda_s \pi_s + \lambda_d \pi_d] = \pi(a \mid y) + \alpha \lambda_c \sum_j w_j [\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})]$. In log space, $\log(\lambda_c u + r) - \log(\lambda_c v + r)$ depends on $r = \lambda_s \pi_s + \lambda_d \pi_d$ unless $u = v$, so the shortcut terms do not cancel. □

Proposition 3. Let $m(a) = \pi(a \mid Q, P)$. CAD prefers a_d over g iff $(1{+}\alpha) \log \frac{m(a_d)}{m(g)} > \alpha \log \frac{\pi_s(a_d)}{\pi_s(g)}$. The left side is bounded below by $(1{+}\alpha) \log \frac{\lambda_d}{m(g)}$ since $m(a_d) \geq \lambda_d$, while the right side tends to $-\infty$ as $\pi_s(a_d) \to 0$. For HGCD, both terms of eq. (1) condition on the same P, so $\lambda_d \pi_d$ enters both and cancels (proposition 2). □

Proposition 4. Under exogeneity and monotonicity, Lemma 2 of Yu et al. [36] gives PNS = $P(A{=}a \mid \mathrm{do}(S)) - P(A{=}a \mid \mathrm{do}(S'))$. With $S = y$ and $S' = y_j^{\mathrm{tot}}$ this is $\Delta_j^{\mathrm{tot}}(a)$. Dividing eq. (1) by α and letting $\alpha \to \infty$ removes the $\pi(a \mid y)/\alpha$ term. □

B Implementation and Reproducibility

Candidate log-probabilities are computed with one prefix forward and a KV-cached pass over the candidate tokens (verified against full recomputation to 2×10^{-5}). Prompts are two-shot, hop-structured; the second CLUTRR shot uses a relation outside the label set to avoid leaking label statistics. All interventions are seeded functions of the example id. The Kaggle notebook embeds the full package and writes every table in this paper.

C Pilot run (pipeline sanity check)

The following tables come from a CPU smoke run with Qwen2.5-0.5B-Instruct on a handful of examples. They verify that the pipeline runs end to end and are **not** evidence for or against any hypothesis.

Estimator	clutrr		musique	
	all	PS=1	all	PS=1
Δ^{naive} (mask hop only)	0.900	1.000	0.802	0.862
Δ^{total} (graph-aware mask)	0.881	1.000	1.000	1.000
DRG path indicator	1.000	1.000	1.000	1.000
$\Delta^{\mathrm{placebo}}$ (content-preserving)	0.469	0.300	0.521	0.538
Δ^{marker} (mask token, content kept)	0.581	0.700	0.469	0.462

Dataset	Hop type	Δ^{total}	Δ^{naive}	$\Delta^{\mathrm{placebo}}$	Δ^{marker}
clutrr	gold-final	0.079	0.079	0.036	0.023
clutrr	gold-nonfinal	0.037	0.011	-0.048	-0.029
clutrr	redundant	-0.072	-0.072	-0.007	-0.018
musique	gold-final	0.682	0.682	-0.077	-0.062
musique	gold-nonfinal	0.423	-0.036	-0.050	-0.012
musique	redundant	-0.112	-0.112	-0.074	-0.018

Method	ID (n=8)	OOD (n=4)	All (n=12)
Direct (no CoT)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
CoT	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
Self-consistency	12.5 [0.0,37.5]	0.0 [0.0,0.0]	8.3 [0.0,25.0]
CAD	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
AdaCAD	37.5 [12.5,75.0]	0.0 [0.0,0.0]	25.0 [0.0,50.0]
CAD-chain	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (naive mask)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (no graph)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (log-space)	37.5 [12.5,75.0]	25.0 [0.0,75.0]	33.3 [8.3,58.3]
HGCD (adaptive α)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD (placebo)	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
HGCD	25.0 [0.0,62.5]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
L1 + CoT	25.0 [0.0,50.0]	0.0 [0.0,0.0]	16.7 [0.0,41.7]
L1 + HGCD	25.0 [0.0,50.0]	0.0 [0.0,0.0]	16.7 [0.0,41.7]

Even at this scale, the position breakdown shows the qualitative pattern predicted by proposition 1 on MuSiQue: naive masking leaves the answer probability of non-final gold hops nearly unchanged, whereas graph-aware masking reduces it substantially. With a few dozen hops this is an illustration, not a test.